AI-Assisted Coding
Lab Assignment - 18.4

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Lab 18: API Integration: Connecting to External Services with Error Handling

Task 1 – Movie Database API Integration

Scenario: You are building a movie recommendation tool that fetches movie details from a public Movie Database API.
Task: Use AI to generate a Python script that connects to a Movie API and retrieves movie information by title.

Prompt: Create a Python script using the requests library to fetch movie details from OMDb API.Requirements: Take movie title as user input, Fetch data from API using API key,Display: Title, Year, IMDb Rating, Plot,Handle errors: invalid movie, API key issues, network errors, empty response,Use clean formatting and functions for better readability.

Code:

ass-18.4.py > get_movie_details

```python
#Create a Python script using the requests library to fetch movie
#details from OMDb API.Requirements: Take movie title as user input, Fetch
#data from API using API key,Display: Title, Year, IMDb Rating, Plot,Handle
#errors: invalid movie, API key issues, network errors, empty response,Use clean
#formatting and functions for better readability.
import requests
API_KEY = "d69f7ff3"
BASE_URL = " http://www.omdbapi.com/"
def get_movie_details(movie_title):
    params = {
        "apikey": API_KEY,
        "t": movie_title
    }
    try:
        response = requests.get(BASE_URL, params=params)
        response.raise_for_status()  # Raise an error for HTTP errors
        data = response.json()
        if data.get("Response") == "False":
            print("Movie not found.")
            return None
        return data
    except requests.RequestException as e:
        print(f"Error fetching movie details: {e}")
        return None
def display_movie_details(movie_data):
    if not movie_data:
        return
    print(f"Title: {movie_data.get('Title', 'N/A')}")
    print(f"Year: {movie_data.get('Year', 'N/A')}")
    print(f"IMDb Rating: {movie_data.get('imdbRating', 'N/A')}")
    print(f"Plot: {movie_data.get('Plot', 'N/A')}")
def main():
    movie_title = input("Enter the movie title: ")
    movie_data = get_movie_details(movie_title)
    display_movie_details(movie_data)
if __name__ == "__main__":
    main()
```

Ln 24, Col 20 Spaces: 4 UTF-8 CRLF {} Python 3.11.4 Python 3.11 (64-bit) Go Live

Output:

```
Enter the movie title: Guardians of the Galaxy Vol. 2
Title: Guardians of the Galaxy Vol. 2
Year: 2017
IMDb Rating: 7.6
Plot: The Guardians struggle to keep together as a team while dealing with their personal family issues, notably Star-Lord's encounter with his father, the ambitious ce
lestial being Ego.
```

Explanation: This script allows users to fetch movie details from the OMDb API by entering a movie title and an API key. It uses the requests library to make an HTTP GET request to the API, retrieves the movie details in JSON format, and displays the title, year, IMDb rating, and plot. The script also includes error handling for invalid movie titles, API key issues, network errors, and empty responses.

Task 2 – Public Transport Schedule API

Scenario: A city transport app needs to fetch bus/train arrival timings from a transport API.

Task: Use AI assistance to integrate a transport schedule API.

Prompt: Create a Python script using the requests library to fetch public transport arrival times from an API. Take stop ID as user input, validate it, and display the next 3 arrivals. Handle errors like invalid stop ID, server issues, timeout, and empty response using try-except. Include at least one retry attempt and keep the output clean and readable.

Code:

```python
import requests
import time
from datetime import datetime

API_URL = "https://api.tfl.gov.uk/StopPoint/{stop_id}/Arrivals"


def fetch_arrival_times(stop_id):
    url = API_URL.format(stop_id=stop_id)

    try:
        response = requests.get(url, timeout=5)
        response.raise_for_status()
        data = response.json()

        # TfL returns a list, not a dict
        if not isinstance(data, list) or len(data) == 0:
            print("⚠️ No arrivals found for this stop.")
            return None

        # Sort by arrival time
        data.sort(key=lambda x: x.get("timeToStation", 999999))

        return data[:3]  # next 3 arrivals

    except requests.exceptions.HTTPError as e:
        print("❌ Invalid Stop ID or API issue:", e)

    except requests.exceptions.ConnectionError:
        print("❌ Network error. Check your internet connection.")
```

```python
def fetch_arrival_times(stop_id):
    except requests.exceptions.Timeout:
        print("❌ Request timed out.")

    except requests.exceptions.RequestException as e:
        print("❌ Unexpected error:", e)

    return None


def format_time(seconds):
    minutes = seconds // 60
    return f"{minutes} min"


def display_arrivals(arrivals):
    if not arrivals:
        print("🚫 No data to display.")
        return

    print("\n📋 Next Bus Arrivals")
    print("-" * 40)

    for bus in arrivals:
        line = bus.get("lineName", "N/A")
        destination = bus.get("destinationName", "Unknown")
        time_to_arrival = format_time(bus.get("timeToStation", 0))

        print(f"🚌 Bus {line} → {destination}")
        print(f"   Arrival in: {time_to_arrival}")
        print("-" * 40)


def main():
    print("📍 Welcome to Smart Bus Arrival Checker")
```

Output:

```python
def main():
    print("📍 Welcome to Smart Bus Arrival Checker")
    stop_id = input("Enter TfL Stop ID (e.g., 490008660N): ").strip()
    if not stop_id:
        print("❌ Stop ID cannot be empty.")
        return
    arrivals = fetch_arrival_times(stop_id)
    if arrivals is None:
        print("🔄 Retrying in 2 seconds...")
        time.sleep(2)
        arrivals = fetch_arrival_times(stop_id)

    display_arrivals(arrivals)
if __name__ == "__main__":
    main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\vempa\OneDrive\Pictures\Desktop\HCL LAB\AI ASSISTANT CODING> & C:/Users/vempa/AppData/Local/Programs/Python/Python311/p
📍 Welcome to Smart Bus Arrival Checker
Enter TfL Stop ID (e.g., 490008660N): 490008660N

📋 Next Bus Arrivals
----------------------------------------
🚌 Bus 88 → Parliament Hill Fields
   Arrival in: 2 min
----------------------------------------
🚌 Bus 88 → Parliament Hill Fields
   Arrival in: 10 min
----------------------------------------
🚌 Bus 88 → Parliament Hill Fields
   Arrival in: 23 min
----------------------------------------
```

Explanation:
The program takes a stop ID as input and fetches arrival data from the TfL API using the requests library. It processes the response, sorts arrivals by nearest time, and displays the next 3 arrivals with details. It also handles errors (timeout, connection, invalid data) and includes a retry mechanism for reliability.

Task 3 – Stock Market Data API

Scenario: You are developing a finance dashboard that retrieves live stock prices.

Task: Use AI to generate a Python script that fetches stock data from a financial API.

Prompt: Create a Python script using the requests library to fetch live stock data from a financial API (such as Alpha Vantage). The program should accept a stock symbol as user input and display the current price, day high, and day low. It must handle errors like invalid stock symbols, API rate limits, and missing fields in the JSON response. Ensure proper validation and safe parsing of the API response using try-except blocks, and display clean, user-friendly output without any crashes.

Code:

```python
import requests

API_KEY = "YOUR_API_KEY"   # Replace with your Alpha Vantage API key
BASE_URL = "https://www.alphavantage.co/query"


def fetch_stock_data(symbol):
    if not symbol.strip():
        print("❌ Stock symbol cannot be empty.")
        return None

    params = {
        "function": "GLOBAL_QUOTE",
        "symbol": symbol.upper().strip(),
        "apikey": API_KEY
    }

    try:
        response = requests.get(BASE_URL, params=params, timeout=5)
        response.raise_for_status()
        data = response.json()

        # Handle API rate limit
        if "Note" in data:
            print("⚠️ API limit reached. Please try again later.")
            return None

        # Handle invalid API key
        if "Error Message" in data:
```

```python
            print("❌ Invalid API key or request.")
            return None

        quote = data.get("Global Quote")

        if not quote:
            print("❌ Invalid stock symbol or no data found.")
            return None

        return quote

    except requests.exceptions.ConnectionError:
        print("❌ Network error. Check your internet connection.")

    except requests.exceptions.Timeout:
        print("❌ Request timed out. Try again.")

    except requests.exceptions.RequestException as e:
        print("❌ Unexpected error:", e)

    return None


def display_stock_data(quote):
    try:
        symbol = quote.get("01. symbol", "N/A")
        price = quote.get("05. price", "N/A")
        high = quote.get("03. high", "N/A")
        low = quote.get("04. low", "N/A")
```

```python
        print("\n📈 Stock Details")
        print("-" * 40)
        print(f"Symbol      : {symbol}")
        print(f"Current Price: ${price}")
        print(f"Day High    : ${high}")
        print(f"Day Low     : ${low}")
        print("-" * 40)

    except Exception:
        print("❌ Error displaying stock data (missing fields).")


def main():
    print("📊 Live Stock Price Checker")

    symbol = input("Enter stock symbol (e.g., AAPL, TSLA): ")

    stock_data = fetch_stock_data(symbol)

    if stock_data:
        display_stock_data(stock_data)
    else:
        print("⚠️ Unable to retrieve stock data.")


if __name__ == "__main__":
    main()
```

Output:

```
📈 Live Stock Price Checker
Enter stock symbol (e.g., AAPL, TSLA): AAPL

📊 Stock Details
----------------------------------------
Symbol      : AAPL
Current Price: $253.7900
Day High    : $255.4800
Day Low     : $247.1010
----------------------------------------
```

Explanation: The script takes a stock symbol as input and fetches its current price, day high, and day low from a financial API. It safely handles errors like invalid symbols, rate limits, and missing data, ensuring clean output without crashes.

Task 4 – Geolocation API (IP Address Lookup)

Scenario: A cybersecurity tool needs to detect the geographical location of a user based on IP address.

Task: Integrate a Geolocation API using AI assistance.

Prompt: Generate a Python script that takes an IP address as input, validates its format, and uses a Geolocation API (like ip-api.com) to fetch country, city, and ISP. Include error handling for invalid IPs, API failures, and JSON parsing issues, and display results clearly.

Code:

```python
import requests
import re

API_URL = "http://ip-api.com/json/"


def is_valid_ip(ip):
    """Validate IPv4 address format"""
    pattern = re.compile(r"^(?:[0-9]{1,3}\.){3}[0-9]{1,3}$")
    if not pattern.match(ip):
        return False

    # Check each octet is between 0-255
    parts = ip.split(".")
    for part in parts:
        if int(part) < 0 or int(part) > 255:
            return False

    return True


def fetch_ip_details(ip):
    """Fetch geolocation data from API"""
    try:
        url = API_URL + ip
        response = requests.get(url, timeout=5)
        response.raise_for_status()

        try:
```

```python
            data = response.json()
        except ValueError:
            print("❌ Error parsing server response (invalid JSON).")
            return None

        # API-specific error handling
        if data.get("status") == "fail":
            print(f"❌ API Error: {data.get('message', 'Invalid IP address')}")
            return None

        return data

    except requests.exceptions.ConnectionError:
        print("❌ Network error. Please check your internet connection.")

    except requests.exceptions.Timeout:
        print("❌ Request timed out. Try again later.")

    except requests.exceptions.HTTPError as e:
        print("❌ HTTP error:", e)

    except requests.exceptions.RequestException as e:
        print("❌ Unexpected error:", e)

    return None


def display_ip_details(data):
    """Display results cleanly"""
    if not data:
        print("⚠️ No data to display.")
        return

    print("\n🌍 IP Geolocation Details")
    print("-" * 40)
    print(f"IP Address : {data.get('query', 'N/A')}")
    print(f"Country    : {data.get('country', 'N/A')}")
    print(f"City       : {data.get('city', 'N/A')}")
    print(f"ISP        : {data.get('isp', 'N/A')}")
    print("-" * 40)


def main():
    print("🌐 IP Geolocation Finder")

    ip = input("Enter an IP address: ").strip()

    if not is_valid_ip(ip):
        print("❌ Invalid IP address format.")
        return

    data = fetch_ip_details(ip)

    if data:
        display_ip_details(data)
    else:
        print("⚠️ Could not retrieve IP details.")
```

Output:

```
if __name__ == "__main__":
    main()

🌐 IP Geolocation Finder
Enter an IP address: 8.8.8.8

🌍 IP Geolocation Details
----------------------------------------
IP Address : 8.8.8.8
Country    : United States
City       : Ashburn
ISP        : Google LLC
----------------------------------------
```

Explanation: This script takes an IP address, checks if it is valid, and then calls a Geolocation API to get details like country, city, and ISP. It also handles errors such as invalid IP format, API failure, and incorrect JSON response to ensure the program runs safely.

Task 5 – Real-Time Application: Payment Gateway Simulation API

Scenario: You are building a test payment integration system that connects to a sandbox payment API.

Task: Use AI to generate a Python script that simulates payment processing.

Prompt: Generate a Python script that simulates payment processing using a sandbox Payment API. It should send a POST request with amount, currency, and payment method, validate inputs, handle errors (invalid details, HTTP errors, network issues), display transaction status, and log failed transactions to a file.

Code:

ass-18.4.py > ...

```python
import requests
import json
from datetime import datetime

# Example sandbox endpoint (mock API)
API_URL = "https://jsonplaceholder.typicode.com/posts"


def validate_payment(amount, currency, method):
    if amount <= 0:
        print("❌ Amount must be greater than 0.")
        return False

    if not currency.isalpha() or len(currency) != 3:
        print("❌ Currency must be a 3-letter code (e.g., USD, INR).")
        return False

    valid_methods = ["card", "upi", "netbanking"]
    if method.lower() not in valid_methods:
        print(f"❌ Invalid payment method. Choose from {valid_methods}")
        return False

    return True


def log_failed_transaction(data, error_message):
    with open("failed_transactions.log", "a") as file:
        log_entry = {
            "timestamp": str(datetime.now()),
            "data": data,
            "error": error_message
        }
        file.write(json.dumps(log_entry) + "\n")


def process_payment(amount, currency, method):
```

valid_methods = ["card", "upi", "netbanking"]

Full name: ass-18.4.validate_payment.valid_methods

ass-18.4.py > ...

```python
def process_payment(amount, currency, method):
    payload = {
        "amount": amount,
        "currency": currency.upper(),
        "payment_method": method.lower()
    }

    try:
        response = requests.post(API_URL, json=payload, timeout=5)
        response.raise_for_status()

        try:
            data = response.json()
        except ValueError:
            print("❌ Invalid JSON response from server.")
            log_failed_transaction(payload, "JSON parsing error")
            return None

        return data

    except requests.exceptions.HTTPError as e:
        print("❌ HTTP error:", e)
        log_failed_transaction(payload, str(e))

    except requests.exceptions.ConnectionError:
        print("❌ Network error. Check your connection.")
        log_failed_transaction(payload, "Connection error")

    except requests.exceptions.Timeout:
        print("❌ Request timed out.")
        log_failed_transaction(payload, "Timeout")

    except requests.exceptions.RequestException as e:
        print("❌ Unexpected error:", e)
        log_failed_transaction(payload, str(e))

    return None
```

```python
def display_result(response_data):
    if not response_data:
        print("⚠️ Payment failed.")
        return

    print("\n💳 Payment Status")
    print("-" * 40)
    print("Status      : SUCCESS ✅")
    print(f"Transaction ID: {response_data.get('id', 'N/A')}")
    print("-" * 40)


def main():
    print("💰 Payment Processing Simulator")

    try:
        amount = float(input("Enter amount: "))
    except ValueError:
        print("❌ Invalid amount.")
        return

    currency = input("Enter currency (e.g., USD, INR): ").strip()
    method = input("Enter payment method (card/upi/netbanking): ").strip()

    if not validate_payment(amount, currency, method):
        return

    response = process_payment(amount, currency, method)
    display_result(response)


if __name__ == "__main__":
    main()
```

Output:

```
💰 Payment Processing Simulator
Enter amount: 500
Enter currency (e.g., USD, INR): INR
Enter payment method (card/upi/netbanking): upi

Enter currency (e.g., USD, INR): INR
Enter payment method (card/upi/netbanking): upi


💳 Payment Status
----------------------------------------
Status      : SUCCESS ✅
Transaction ID: 101
----------------------------------------
```

Explanation:

This script sends payment details (amount, currency, method) to a sandbox API, checks if inputs are valid, and processes the transaction. It handles errors like invalid data, server issues, or network failures, shows the payment status, and logs failed transactions to a file.